

BLUESTOCK FINTECH

Mutual Fund Analytics

Capstone Project I

Final Project Report

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Data Engineering • Exploratory Data Analysis • Performance Analytics • Business Intelligence

1. Executive Summary

This capstone project develops an end-to-end mutual fund analytics workflow for Bluestock Fintech. The solution combines data ingestion and cleaning, relational data modelling, exploratory analysis, investment performance measurement, advanced analytics, and business-intelligence visualization. The objective is to transform heterogeneous mutual-fund datasets into a structured analytical foundation that can support portfolio research, investor segmentation, risk analysis, and management reporting.

The supplied data package contains fund master information, daily NAV history, fund-house AUM, monthly SIP inflows, industry folio counts, scheme-level performance measures, investor transactions, portfolio holdings, and benchmark-index observations. The data was processed using Python and Pandas, standardized, validated, and loaded into SQLite using a defined relational schema.

The analytical layer examines NAV evolution, AUM concentration, SIP activity, investor demographics, geographic participation, folio growth, return correlations, portfolio sector exposure, and risk measures. The project also defines a path toward advanced analytics including Value at Risk, cohort analysis, and scheme recommendation logic.

Important data note: the project specification references a category-inflow dataset numbered 05, but that dataset was not present in the supplied CSV package. The analysis therefore does not fabricate missing observations; where category inflows are discussed, the limitation is explicitly documented.

2. Problem Statement and Objectives

Mutual fund businesses generate large volumes of time-series, scheme, investor, transaction, and portfolio data. Without a consistent analytical pipeline, these datasets are difficult to validate, compare, and use for decision support. The project addresses this problem by creating a reproducible workflow from raw CSV data through analytics and dashboard-ready outputs.

Project objectives

- Build an automated ETL pipeline for the supplied mutual-fund datasets.
- Standardize dates, column names, numeric fields, and text values.
- Create a normalized SQLite analytical database with appropriate fact and dimension tables.
- Perform exploratory data analysis using Python visualization libraries.
- Calculate return, volatility, risk-adjusted performance, and drawdown measures.
- Analyze investor behavior, portfolio holdings, fund-house AUM, and benchmark relationships.
- Create an interactive business-intelligence dashboard with multiple analytical views.
- Produce professional documentation and presentation material for the complete solution.

3. Data Sources

The analysis uses the supplied mutual-fund data package. The datasets cover scheme metadata, daily market observations, fund-house assets, SIP activity, investor transactions, holdings, and benchmark indices.

Dataset	Purpose	Observed scale
01 Fund Master	Scheme metadata and fund-house information	40 schemes
02 NAV History	Historical daily NAV observations	46,000 rows
03 AUM by Fund House	Fund-house AUM observations	90 rows
04 Monthly SIP Inflows	Monthly SIP contribution analysis	48 months
06 Industry Folio Count	Industry folio growth	21 observations
07 Scheme Performance	Return and risk measures	40 schemes
08 Investor Transactions	Investor activity and segmentation	32,778 transactions
09 Portfolio Holdings	Scheme-level portfolio composition	322 holdings
10 Benchmark Indices	Benchmark time series	8,050 rows

The NAV dataset covers 40 schemes across 1,150 trading dates from January 2022 through May 2026. The investor transaction dataset contains 5,000 unique investors and records SIP, lumpsum, and redemption activity.

4. ETL Design and Data Engineering

4.1 Pipeline Architecture

Raw CSV files → discovery → column normalization → type conversion → date standardization → validation → processed CSV files → SQLite database → analytics/dashboard.

4.2 Processing steps

- Dataset discovery uses pathlib rather than machine-specific relative-path assumptions.
- Column names are normalized to lower-case, underscore-separated identifiers.
- Date fields are parsed into consistent datetime representations.
- Numeric-like fields are converted using coercion so malformed observations can be identified.
- String values are stripped of leading and trailing whitespace.
- The pipeline checks for duplicate records, missing datasets, missing cells, and orphan AMFI codes.
- The database is created from a SQL schema and populated programmatically.
- A data-quality report is generated to document validation results.

4.3 Database design

The relational model separates scheme-level descriptive information from time-series and transactional facts. The principal dimension table is `dim_fund`. Fact tables store NAV, AUM, SIP, folio, performance, investor transactions, portfolio holdings, and benchmark observations. Indexes are used on frequently queried keys such as AMFI codes and dates.

5. Exploratory Data Analysis

EDA was designed to cover market performance, fund-house scale, investor behavior, geography, industry participation, correlations, and portfolio composition. Interactive Plotly charts are used for selected time-series views, while Seaborn and Matplotlib support static analytical charts.

5.1 NAV trend analysis

The indexed-NAV analysis normalizes each scheme's starting observation to 100, allowing schemes with different absolute NAV levels to be compared on a common growth scale. The analysis highlights the evolution of schemes over the 2022–2026 observation window and separates the broad 2023 and 2024 periods for interpretation.

5.2 AUM by fund house

Fund-house AUM was aggregated by year to examine scale and concentration. This provides a management-level view of how assets are distributed across AMCs and how that distribution changes over time.

5.3 SIP activity

Monthly SIP inflows were analyzed across January 2022 to December 2025. The time series is useful for identifying growth phases, seasonal movement, and changes in systematic-investment participation.

5.4 Investor demographics

- Age-group distributions provide a demographic view of investor participation.
- SIP amounts can be compared across age groups using distributional charts such as box plots.
- Gender segmentation provides a basic view of participation composition.

5.5 Geographic participation

State-level SIP activity and T30/B30 segmentation provide a geographic view of investor participation. These measures can help identify concentration and expansion opportunities.

5.6 Folio growth

Industry folio counts were plotted as a time series to show the expansion of investor accounts over the supplied period. Milestone annotations make long-term growth easier to interpret.

6. Additional EDA Findings

6.1 NAV return correlation

A correlation matrix across a selected set of schemes provides a compact view of co-movement. Higher positive correlations suggest that schemes may respond similarly to common market factors, while lower correlations can indicate greater diversification potential.

6.2 Portfolio sector allocation

Portfolio holdings were aggregated by sector to understand where scheme portfolios are concentrated. A sector allocation chart translates individual holdings into a more interpretable exposure profile.

6.3 Risk and return views

Scheme-level performance data includes return, benchmark return, alpha, beta, Sharpe ratio, Sortino ratio, standard deviation, maximum drawdown, AUM, expense ratio, Morningstar rating, and risk grade. These fields enable multi-dimensional comparison rather than relying on return alone.

6.4 Ten key analytical findings

- The dataset contains 40 distinct mutual-fund schemes across 10 fund houses.
- Daily NAV history provides a sufficiently long time series for multi-year trend analysis.
- Fund-house AUM analysis reveals material differences in asset scale across AMCs.
- Monthly SIP inflows provide a clear time-series view of systematic-investment activity.
- Investor transactions show meaningful differences between SIP, lumpsum, and redemption behavior.
- Investor records support segmentation by age, gender, income, state, city tier, and payment mode.
- Folio counts provide a separate measure of participation growth from transaction volumes.
- Scheme NAV returns are correlated to varying degrees, reflecting shared and scheme-specific market exposure.
- Portfolio holdings allow sector-level concentration analysis.
- The combined dataset supports an integrated view of fund performance, investor behavior, risk, and portfolio composition.

7. Performance Analytics

Performance analytics combines absolute and risk-adjusted measures. The goal is to evaluate how much return a scheme generated and how that return relates to volatility, market exposure, downside risk, and a benchmark.

7.1 CAGR

Compound Annual Growth Rate converts cumulative growth over a period into an annualized rate. For a start value V_0 , end value V_n , and period of n years: **$CAGR = (V_n / V_0)^{(1/n)} - 1$** .

7.2 Annualized return

For trading-day based return series, annualization uses the number of trading observations. The project follows the stated requirement to use 252 trading days when annualizing daily performance measures.

7.3 Volatility

Annualized volatility measures the dispersion of periodic returns. Daily standard deviation can be annualized using the square root of 252.

7.4 Sharpe ratio

The Sharpe ratio measures excess return relative to total volatility. A conventional formulation is **$(\text{portfolio return} - \text{risk-free rate}) / \text{portfolio volatility}$** .

7.5 Beta and Alpha

Beta measures sensitivity to benchmark movements. Alpha measures the return unexplained by the benchmark after accounting for the assumed model inputs.

7.6 Sortino ratio

The Sortino ratio focuses on downside deviation rather than total volatility, making it useful when the analytical objective emphasizes harmful return variation.

8. Risk and Advanced Analytics

8.1 Maximum drawdown

Maximum drawdown measures the largest peak-to-trough decline in a value series. It is particularly useful for communicating the historical downside experience of an investment.

8.2 Value at Risk

Value at Risk estimates a loss threshold for a selected confidence level over a defined horizon. The implementation must clearly identify the chosen historical or parametric method, confidence level, and return frequency.

8.3 Cohort analysis

Investor cohorts can be formed using attributes such as age group, income group, city tier, transaction type, or investment period. Comparing transaction volume and value across cohorts can reveal behavioral differences.

8.4 Recommendation logic

A scheme recommendation layer can combine investor profile characteristics with scheme-level risk and return attributes. The appropriate implementation is a transparent scoring or filtering mechanism rather than an opaque prediction model, so that the factors influencing a recommendation remain auditable.

9. Interactive Dashboard

The Power BI dashboard is intended to provide an executive-to-detail analytical flow. Dashboard design should allow users to move from overall market and fund-house context to scheme performance, investor behavior, and risk.

Recommended dashboard page structure

- Executive Overview — AUM, SIP inflows, scheme count, and headline KPIs.
- Fund Performance — NAV trends, return measures, benchmark comparison, and scheme selectors.
- Investor Analytics — transaction mix, age, gender, income, geography, and city-tier segmentation.
- Risk & Portfolio — volatility, drawdown, risk-adjusted metrics, holdings, and sector exposure.

Each page should contain at least two functional slicers or filters. Screenshots of the completed dashboard should be inserted into the final version of this report before submission.

10. Limitations

- The analysis is constrained by the supplied dataset package and its observation periods.
- The category-inflow dataset referenced in the specification was not included in the supplied files.
- Investor transaction records cover a defined historical period and should not be interpreted as a complete investor population.
- Historical performance metrics describe historical observations and do not establish future investment outcomes.
- SQLite is appropriate for this capstone analytical workflow but is not a substitute for a production-scale data warehouse.
- Some dashboard and advanced-analytics outputs depend on successful execution of the corresponding notebooks/scripts and refresh configuration.

11. Recommendations and Future Enhancements

- Automate scheduled NAV ingestion from a reliable external data source.
- Add a production-grade database or warehouse for larger data volumes.
- Implement automated data-quality monitoring and alerting.
- Add live dashboard refresh and centralized metric definitions.
- Expand investor cohort analysis with retention and behavioral lifecycle measures.
- Add Monte Carlo simulation and Markowitz portfolio optimization as optional advanced modules.
- Automate management reporting through scheduled PDF or HTML report generation.

12. Conclusion

The Bluestock Mutual Fund Analytics capstone establishes a complete analytical workflow that connects data engineering with investment analytics and business intelligence. The project begins with heterogeneous CSV datasets, applies repeatable ETL and validation procedures, stores the resulting information in a structured SQLite database, and exposes the data through EDA, performance metrics, advanced analysis, and dashboard views.

The principal value of the solution is its reproducibility: the same data-processing logic can be rerun when the underlying datasets are refreshed. The separation between ingestion, storage, analysis, and visualization also provides a foundation for future automation and productionization.

The completed project demonstrates practical application of Python, Pandas, SQL, statistical analysis, financial performance measurement, visualization, and Power BI in a single end-to-end workflow.

Appendix A — Core Technologies

- Python 3.x
- Pandas and NumPy
- Matplotlib, Seaborn, and Plotly
- SQLite and SQL
- Jupyter Notebook
- Power BI
- Git and GitHub

Appendix B — Core Project Artifacts

- ETL pipeline script
- SQLite schema and analytical queries
- EDA notebook and exported charts
- Performance analytics notebook and metric outputs
- Advanced analytics notebook
- Power BI dashboard
- Final report and presentation